

JTF003-EC Results

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Results

Data Processing and Exclusions

Exclusion criteria applied as stated in the [preregistration](#):

We will exclude participants who did not complete the study, failed an attention check question, failed to pass the Captcha test (i.e., their Captcha score is equal to or lower than 0.5), or are not permanent residents of the geopolitical region where data collection is taking place.

Table 1: Each line indicates how many participants failed each criterion (applied sequentially).

Exclusion Criterion	Number Excluded	Fraction Excluded	Percentage Excluded
Finished == 1	6528	0.2710176	27.101756
Q_RecaptchaScore >= 0.5	937	0.0389007	3.890065
attention check == 0	551	0.0228754	2.287541
Country_Res_YN == Yes	789	0.0327563	3.275626
Total	8016	0.3327936	33.279362

Table 2: Demographic Information of Participants

Demographic	Value
Average age (Std.)	26.27 (10.79)
Age Range	18 to 99
N. males	4685
N. female	10209
N. non binary	141
N. prefer not to respond	152
N. prefer to self-describe	58

Table 3: Demographic Information of Participants by Cultural Classification

Demographic	Individualistic	Collectivistic
N participants	10525	4720
Average age (SD)	26.91 (11.48)	24.86 (8.89)
Age range	18 to 99	18 to 97
N males	3101	1584
N females	7181	3028
N non-binary	119	22
N prefer not to respond	89	63
N prefer to self-describe	35	23

Table 4: Level of education by cultural group.

Education level	Collectivistic, N (%)	Individualistic, N (%)
Less than a high school or secondary school degree (or equivalent)	34 (0.7%)	119 (1.1%)
High school or secondary school degree (or equivalent)	2,207 (46.8%)	6,447 (61.3%)
Bachelor's degree (or equivalent)	1,833 (38.8%)	2,531 (24.0%)
Postgraduate degree (MA, PhD, JD, etc.)	552 (11.7%)	1,330 (12.6%)
Prefer not to respond	94 (2.0%)	98 (0.9%)

Use of external resources

Table 5: Percentage and number of responses reporting use of external resources, split by problem and condition.

problem	Baseline	Feedback	Justification	Feedback + justification
Problem 1	6.64% (N = 258)	6.82% (N = 265)	7.70% (N = 286)	7.07% (N = 266)
Problem 2	6.95% (N = 270)	6.41% (N = 249)	8.24% (N = 306)	6.73% (N = 253)
Problem 3	10.24% (N = 398)	9.60% (N = 373)	11.69% (N = 434)	11.30% (N = 425)
Problem 4	8.98% (N = 349)	8.70% (N = 338)	11.23% (N = 417)	10.37% (N = 390)
Problem 5	9.57% (N = 372)	9.91% (N = 385)	11.40% (N = 423)	11.75% (N = 442)
Problem 6	10.86% (N = 422)	9.27% (N = 360)	12.93% (N = 480)	12.58% (N = 473)

Familiarity

Table 6: Percentage and number of participants reporting familiarity, split by problem and condition.

problem	Baseline	Feedback	Justification	Feedback + justification
Problem 1	50.19% (N = 1951)	52.72% (N = 2048)	52.24% (N = 1939)	55.04% (N = 2070)
Problem 2	45.02% (N = 1750)	46.51% (N = 1807)	45.37% (N = 1684)	46.82% (N = 1761)
Problem 3	16.03% (N = 623)	14.47% (N = 562)	16.92% (N = 628)	16.35% (N = 615)
Problem 4	31.31% (N = 1217)	31.02% (N = 1205)	31.84% (N = 1182)	31.67% (N = 1191)
Problem 5	26.19% (N = 1018)	25.41% (N = 987)	27.45% (N = 1019)	27.12% (N = 1020)
Problem 6	23.10% (N = 898)	24.20% (N = 940)	24.52% (N = 910)	24.75% (N = 931)

Language background of participants

Response	N	%
Fluency among non-native speakers		
Prefer not to respond	21	1.0
Beginner	44	2.1
Intermediate	397	19.2
Advanced	911	44.2
Expert/Proficient	690	33.4
Native speaker of the study language		
Yes	12888	85.5
No	2064	13.7
Prefer not to respond	120	0.8

Fluency was assessed only among participants who reported that the study language was not their native language.

Language background

Confirmatory Analyses

Table 7: Mean accuracy by CRT problem and phase (here 'intuitive' is phase 1 and 'reflective' is phase 2). Computed after applying exclusion criteria based on familiarity and use of external resource.

	justification	feedback	problem	intuitive	reflective
J		F	1	0.20	0.75
J		F	2	0.43	0.85
J		F	3	0.10	0.53
J		F	4	0.23	0.76
J		F	5	0.08	0.47
J		F	6	0.22	0.62
J		NF	1	0.19	0.49
J		NF	2	0.42	0.74
J		NF	3	0.10	0.49
J		NF	4	0.25	0.57
J		NF	5	0.09	0.35
J		NF	6	0.22	0.48
NJ		F	1	0.21	0.75
NJ		F	2	0.40	0.81
NJ		F	3	0.11	0.38
NJ		F	4	0.27	0.71
NJ		F	5	0.09	0.35
NJ		F	6	0.24	0.54
NJ		NF	1	0.18	0.44
NJ		NF	2	0.41	0.67
NJ		NF	3	0.11	0.33
NJ		NF	4	0.25	0.50
NJ		NF	5	0.10	0.29
NJ		NF	6	0.23	0.39

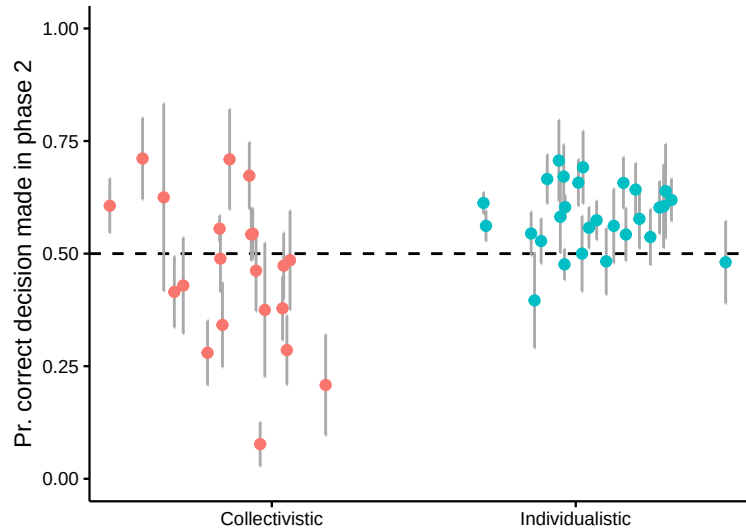


Figure 1: Probability that a correct decision is made in phase 2 rather than in phase 1 (H1 variable), split by country and culture. Error bars are standard errors between participants in each country.

Models for H1a and H1b(Correction during deliberation)

Table 8: Model for H2a and H3a.

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	0.146	0.071	[0.007, 0.285]	.039
feedback	0.334	0.043	[0.249, 0.418]	< .001
justification	0.258	0.040	[0.180, 0.336]	< .001
feedback:justification	-0.039	0.055	[-0.147, 0.068]	.470
SD (Country: (Intercept))	0.451			
SD (Country: feedback)	0.083			
SD (Country: justification)	0.000			
SD (Country: feedback:justification)	0.000			
N	24223			
N Country	53			

Table 9: Model for H1b; individualistic countries.

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	0.348	0.056	[0.239, 0.458]	< .001
feedback	0.328	0.050	[0.230, 0.426]	< .001
justification	0.249	0.046	[0.159, 0.339]	< .001
feedback:justification	-0.003	0.063	[-0.126, 0.121]	.968
SD (Country: (Intercept))	0.235			
SD (Country: feedback)	0.081			
SD (Country: justification)	0.000			
SD (Country: feedback:justification)	0.000			

Predictor	Estimate	SE	95% CI	p-value
N	18276			
N Country	31			

Table 10: Model for H1b; collectivistic countries.

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	-0.160	0.133	[-0.421, 0.101]	.229
feedback	0.358	0.077	[0.207, 0.508]	< .001
justification	0.280	0.080	[0.124, 0.435]	< .001
feedback:justification	-0.142	0.108	[-0.354, 0.070]	.190
SD (Country: (Intercept))	0.542			
SD (Country: feedback)	0.000			
SD (Country: justification)	0.000			
SD (Country: feedback:justification)	0.000			
N	5947			
N Country	22			

Models for H2 and H3 (Effects of accuracy and justification)

Table 11: Model for H2a and H3a.

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	-0.563	0.070	[-0.699, -0.427]	< .001
feedback	0.594	0.028	[0.540, 0.649]	< .001
justification	0.387	0.035	[0.318, 0.456]	< .001
feedback:justification	-0.041	0.047	[-0.134, 0.051]	.382
SD (Country: (Intercept))	0.476			
SD (Country: feedback)	0.040			
SD (Country: justification)	0.128			
SD (Country: feedback:justification)	0.158			
N	48038			
N Country	53			

Table 12: Model for H2b and H3b; individualistic countries.

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	-0.376	0.068	[-0.509, -0.243]	< .001
feedback	0.597	0.031	[0.536, 0.658]	< .001
justification	0.407	0.040	[0.329, 0.485]	< .001
feedback:justification	0.012	0.047	[-0.080, 0.104]	.791
SD (Country: (Intercept))	0.349			
SD (Country: feedback)	0.000			
SD (Country: justification)	0.107			
SD (Country: feedback:justification)	0.054			
N	33989			
N Country	31			

Table 13: Model for H2b and H3b; collectivistic countries.

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	-0.835	0.115	[-1.059, -0.610]	< .001
feedback	0.600	0.057	[0.488, 0.712]	< .001
justification	0.363	0.068	[0.230, 0.495]	< .001
feedback:justification	-0.161	0.094	[-0.345, 0.023]	.086
SD (Country: (Intercept))	0.487			
SD (Country: feedback)	0.063			
SD (Country: justification)	0.156			
SD (Country: feedback:justification)	0.217			
N	14049			
N Country	22			

Confirmatory hypothesis testing summary

Table 14: Summary of preregistered hypothesis tests after including by-problem random intercepts

Hypothesis	Prediction	Supported	p-value(s)
H1a	Less than half of correct answers will arise from error correction during deliberation	No	.981
H1b	Baseline performance is below 50% in both cultural groups	No	Individualistic = 1.000; Collectivistic = .115
H2a	Feedback increases correction overall	Yes	< .001
H2b	Feedback increases correction in both cultural groups	Yes	Individualistic = < .001; Collectivistic = < .001
H3a	Justification increases correction overall	Yes	< .001
H3b	Justification increases correction in both cultural groups	Yes	Individualistic = < .001; Collectivistic = < .001

Robustness Checks

Random Effects for CRT Problem

Table 15: Model for H2a and H3a (model with random intercepts for CRT problem).

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	0.238	0.203	[-0.160, 0.637]	.242
feedback	0.340	0.045	[0.252, 0.428]	< .001
justification	0.250	0.041	[0.170, 0.330]	< .001
feedback:justification	-0.049	0.057	[-0.161, 0.063]	.393
SD (Country: (Intercept))	0.471			
SD (Country: feedback)	0.103			
SD (Country: justification)	0.000			
SD (Country: feedback:justification)	0.059			
SD (problem: (Intercept))	0.465			
N	24223			
N Country	53			
N problem	6			

Table 16: Model for H1b (model with random intercepts for CRT problem); individualistic countries.

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	0.462	0.213	[0.045, 0.880]	.030
feedback	0.325	0.054	[0.220, 0.430]	< .001
justification	0.237	0.047	[0.144, 0.329]	< .001
feedback:justification	-0.008	0.067	[-0.139, 0.123]	.904
SD (Country: (Intercept))	0.259			
SD (Country: feedback)	0.112			
SD (Country: justification)	0.000			
SD (Country: feedback:justification)	0.061			
SD (problem: (Intercept))	0.501			
N	18276			
N Country	31			
N problem	6			

Table 17: Model for H1b (model with random intercepts for CRT problem); collectivistic countries.

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	-0.105	0.207	[-0.511, 0.301]	.611
feedback	0.370	0.082	[0.209, 0.531]	< .001
justification	0.282	0.081	[0.122, 0.441]	< .001
feedback:justification	-0.153	0.111	[-0.369, 0.064]	.167
SD (Country: (Intercept))	0.562			
SD (Country: feedback)	0.082			
SD (Country: justification)	0.000			
SD (Country: feedback:justification)	0.000			
SD (problem: (Intercept))	0.378			
N	5947			
N Country	22			
N problem	6			

Table 18: Model for H2a and H3a (model with random intercepts for CRT problem).

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	-0.539	0.221	[-0.972, -0.105]	.015
feedback	0.635	0.030	[0.576, 0.693]	< .001
justification	0.415	0.037	[0.342, 0.489]	< .001
feedback:justification	-0.050	0.050	[-0.147, 0.048]	.317
SD (Country: (Intercept))	0.511			
SD (Country: feedback)	0.058			
SD (Country: justification)	0.145			
SD (Country: feedback:justification)	0.175			
SD (problem: (Intercept))	0.519			
N	48038			
N Country	53			
N problem	6			

Table 19: Model for H2b and H3b (model with random intercepts for CRT problem); individualistic countries.

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	-0.340	0.237	[-0.804, 0.123]	.150
feedback	0.643	0.032	[0.580, 0.706]	< .001
justification	0.440	0.042	[0.357, 0.523]	< .001
feedback:justification	0.006	0.049	[-0.090, 0.101]	.910
SD (Country: (Intercept))	0.387			
SD (Country: feedback)	0.000			
SD (Country: justification)	0.122			
SD (Country: feedback:justification)	0.058			
SD (problem: (Intercept))	0.553			
N	33989			
N Country	31			
N problem	6			

Table 20: Model for H2b and H3b (model with random intercepts for CRT problem); collectivistic countries.

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	-0.816	0.235	[-1.277, -0.355]	< .001
feedback	0.637	0.060	[0.520, 0.754]	< .001
justification	0.385	0.071	[0.246, 0.525]	< .001
feedback:justification	-0.168	0.097	[-0.359, 0.023]	.085
SD (Country: (Intercept))	0.518			
SD (Country: feedback)	0.077			
SD (Country: justification)	0.170			
SD (Country: feedback:justification)	0.231			
SD (problem: (Intercept))	0.494			
N	14049			
N Country	22			
N problem	6			

Hypothesis testing summary

Table 21: Summary of preregistered hypothesis tests after including by-problem random intercepts

Hypothesis	Prediction	Supported	p-value(s)
H1a	Less than half of correct answers will arise from error correction during deliberation	No	.879
H1b	Baseline performance is below 50% in both cultural groups	No	Individualistic = .985; Collectivistic = .306
H2a	Feedback increases correction overall	Yes	< .001
H2b	Feedback increases correction in both cultural groups	Yes	Individualistic = < .001; Collectivistic = < .001
H3a	Justification increases correction overall	Yes	< .001
H3b	Justification increases correction in both cultural groups	Yes	Individualistic = < .001; Collectivistic = < .001

Random Effects for Participating Site

Table 22: Model for H2a and H3a (model with random intercepts for participating site).

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	0.170	0.069	[0.034, 0.305]	.014
feedback	0.333	0.044	[0.247, 0.419]	< .001
justification	0.255	0.040	[0.176, 0.334]	< .001
feedback:justification	-0.030	0.055	[-0.139, 0.078]	.585
SD (LabID: (Intercept))	0.308			
SD (Country: feedback:justification)	0.000			
SD (Country: justification)	0.000			
SD (Country: feedback)	0.096			
SD (Country: (Intercept))	0.350			
N	23957			
N LabID	135			
N Country	52			

Table 23: Model for H1b (model with random intercepts for participating site); individualistic countries.

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	0.344	0.052	[0.243, 0.445]	< .001
feedback	0.328	0.051	[0.228, 0.428]	< .001
justification	0.244	0.047	[0.153, 0.335]	< .001
feedback:justification	0.015	0.064	[-0.111, 0.141]	.811
SD (LabID: (Intercept))	0.315			
SD (Country: feedback:justification)	0.000			
SD (Country: justification)	0.000			
SD (Country: feedback)	0.098			
SD (Country: (Intercept))	0.074			
N	18017			
N LabID	93			
N Country	30			

Table 24: Model for H1b (model with random intercepts for participating site); collectivistic countries.

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	-0.150	0.136	[-0.417, 0.117]	.270
feedback	0.357	0.077	[0.206, 0.509]	< .001
justification	0.281	0.080	[0.124, 0.437]	< .001
feedback:justification	-0.150	0.109	[-0.363, 0.063]	.168
SD (LabID: (Intercept))	0.221			
SD (Country: feedback:justification)	0.000			
SD (Country: justification)	0.000			
SD (Country: feedback)	0.000			
SD (Country: (Intercept))	0.515			
N	5940			
N LabID	43			
N Country	22			

Table 25: Model for H2a and H3a (model with random intercepts for participating site).

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	-0.535	0.065	[-0.663, -0.407]	< .001
feedback	0.601	0.027	[0.548, 0.653]	< .001
justification	0.393	0.035	[0.324, 0.462]	< .001
feedback:justification	-0.036	0.047	[-0.129, 0.056]	.441
SD (LabID: (Intercept))	0.374			
SD (Country: feedback:justification)	0.155			
SD (Country: justification)	0.124			
SD (Country: feedback)	0.008			
SD (Country: (Intercept))	0.323			
N	47632			
N LabID	137			
N Country	52			

Table 26: Model for H2b and H3b (model with random intercepts for participating site); individualistic countries.

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	-0.379	0.069	[-0.515, -0.244]	< .001
feedback	0.602	0.032	[0.540, 0.664]	< .001
justification	0.414	0.039	[0.337, 0.491]	< .001
feedback:justification	0.022	0.047	[-0.070, 0.115]	.632
SD (LabID: (Intercept))	0.320			
SD (Country: feedback:justification)	0.041			
SD (Country: justification)	0.098			
SD (Country: feedback)	0.000			
SD (Country: (Intercept))	0.268			
N	33604			
N LabID	95			
N Country	30			

Table 27: Model for H2b and H3b (model with random intercepts for participating site); collectivistic countries.

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	-0.774	0.111	[-0.991, -0.557]	< .001
feedback	0.597	0.050	[0.499, 0.695]	< .001
justification	0.364	0.066	[0.235, 0.494]	< .001
feedback:justification	-0.161	0.091	[-0.340, 0.018]	.078
SD (LabID: (Intercept))	0.483			
SD (Country: feedback:justification)	0.213			
SD (Country: justification)	0.148			
SD (Country: feedback)	0.000			
SD (Country: (Intercept))	0.225			
N	14028			
N LabID	43			
N Country	22			

Hypothesis testing summary

Table 28: Summary of preregistered hypothesis tests after including by-problem random intercepts

Hypothesis	Prediction	Supported	p-value(s)
H1a	Less than half of correct answers will arise from error correction during deliberation	No	.993
H1b	Baseline performance is below 50% in both cultural groups	No	Individualistic = 1.000; Collectivistic = .135
H2a	Feedback increases correction overall	Yes	< .001
H2b	Feedback increases correction in both cultural groups	Yes	Individualistic = < .001; Collectivistic = < .001
H3a	Justification increases correction overall	Yes	< .001
H3b	Justification increases correction in both cultural groups	Yes	Individualistic = < .001; Collectivistic = < .001

Exploratory Analyses

Scales reliability and measurement invariance

Note: for these analyses I have included only countries with $N \geq 100$.

Intellectual humility

Table 29: Internal consistency of the Intellectual Humility scale by country (only countries with $N > 100$ shown).

Country	N	alpha
United States	1672	0.64
Turkey	1468	0.65
Spain	1446	0.64
Italy	896	0.57
United Kingdom	842	0.67
Poland	452	0.62
Philippines	414	0.68
Australia	410	0.66
Germany	373	0.65
Colombia	372	0.67
Malaysia	349	0.62
Bulgaria	348	0.64
Austria	287	0.62
Canada	278	0.64
Netherlands	276	0.67
Greece	273	0.71
India	268	0.71
Pakistan	257	0.64
Slovakia	236	0.69
Hungary	226	0.64
Serbia	220	0.66
Portugal	219	0.67
Russia	201	0.69
Chile	197	0.78
North Macedonia	186	0.62
Egypt	183	0.52
Singapore	176	0.73
Kuwait	168	0.71
France	166	0.56
Czech Republic	158	0.67
Sierra Leone	146	0.71
Nigeria	144	0.67
Kenya	143	0.74
China	136	0.57
South Africa	133	0.72
Indonesia	123	0.64
Mexico	122	0.68
Ghana	110	0.83
Ethiopia	109	0.63
Cameroon	108	0.84

Country	N	alpha
New Zealand	104	0.71

Table 30: Measurement invariance analysis for the Intellectual Humility scale

Model	CFI	TLI	RMSEA	SRMR	Δ CFI	Δ RMSEA
Configural	0.973	0.943	0.050	0.029	NA	NA
Metric	0.956	0.930	0.056	0.046	-0.017	0.005
Scalar	0.924	0.905	0.065	0.062	-0.032	0.009

As an exploratory test of cross-country measurement invariance, we fitted multi-group CFAs in which the nine items loaded on a substantive intellectual-humility (IH) factor while separate positive- and negative-wording method factors captured the method effects identified in the original scale validation. The IH factor was specified as independent of the wording factors, while the two method factors were allowed to covary. The configural model showed good fit (CFI = .973, RMSEA = .050, SRMR = .029). We then constrained only the substantive IH loadings to equality across countries, leaving wording-factor loadings free; this produced a modest deterioration in fit (Δ CFI = -.017, Δ RMSEA = .005, Δ SRMR = .017). Adding equality constraints on item intercepts produced a larger deterioration (Δ CFI = -.032, Δ RMSEA = .009, Δ SRMR = .016). Thus, using the conventional Δ CFI $\leq$.01 criterion, strict metric and scalar invariance were not supported, although departures from metric invariance were relatively modest. These results suggest that the scale captures a broadly similar construct across countries, but some caution is warranted.

Actively open-minded thinking scale

Table 31: Measurement invariance analysis for the Actively Open-Minded Thinking scale

Model	CFI	TLI	RMSEA	SRMR	Δ CFI	Δ RMSEA	Δ SRMR
Configural	0.923	0.894	0.051	0.044	NA	NA	NA
Metric	0.889	0.872	0.056	0.067	-0.034	0.005	0.023
Scalar	0.750	0.752	0.079	0.091	-0.139	0.022	0.024

The AOT scale showed mixed evidence of configural fit (CFI = .923, RMSEA = .051, SRMR = .044). When equality constraints were imposed only on the substantive AOT loadings, while allowing loadings on the negative-wording method factor to vary across countries, model fit deteriorated moderately (Δ CFI = -.034, Δ RMSEA = .005, Δ SRMR = .023), indicating departures from full metric invariance. Imposing equality constraints on item intercepts produced a much larger deterioration in fit (Δ CFI = -.139, Δ RMSEA = .022, Δ SRMR = .024), providing clear evidence against scalar invariance.

Individualistic and collectivistic cultural tendencies scale

The original 1998 paper only reported that an EFA-supported four-factor structure, followed by convergent/discriminant validity analyses; they did not do an extensive four-factor CFA structure validation of the 16-items scale.

NOTE: post-fitting diagnostic check (`lavInspect(fit, "post.check")`) was failing for these model, in particular Egypt, Kuwait and Ghana (with N ranging from 110 to 183) had negative eigenvalues, indicating non-positive defined covariance matrices. Therefore I have increased the threshold for being included in the analysis from 100 to 200 responses (considering also that this is a more complex 4 factor model).

Table 32: Measurement invariance analysis for the Individualism–Collectivism scale

Model	CFI	TLI	RMSEA	SRMR	Δ CFI	Δ RMSEA	Δ SRMR
Configural	0.833	0.796	0.075	0.074	NA	NA	NA
Metric	0.816	0.798	0.075	0.081	-0.018	0.000	0.007
Scalar	0.625	0.627	0.102	0.104	-0.191	0.027	0.023

For the Individualism–Collectivism scale, measurement invariance was examined using the proposed four-factor structure (horizontal individualism, vertical individualism, horizontal collectivism, and vertical collectivism). Because models including smaller country samples produced inadmissible solutions, analyses were restricted to countries with at least 200 participants. The configural model showed mixed absolute fit ($CFI = .833$, $RMSEA = .075$, $SRMR = .074$), providing only moderate support for the four-factor structure across countries. Constraining factor loadings to equality resulted in a relatively small deterioration in fit ($\Delta CFI = -.018$, $\Delta RMSEA = .000$, $\Delta SRMR = .007$); thus, strict metric invariance was not supported according to the conventional $\Delta CFI \leq .01$ criterion, although the results were broadly consistent with approximate metric comparability. In contrast, imposing equality constraints on item intercepts produced a substantial deterioration in fit ($\Delta CFI = -.191$, $\Delta RMSEA = .027$, $\Delta SRMR = .023$), providing clear evidence against scalar invariance. Accordingly, associations involving the four dimensions may be reasonably comparable across countries, but cross-country comparisons of their observed mean levels should be interpreted with considerable caution.

Individual Differences and Corrective Behaviour (pre-registered analyses)

Table 33: Correlations between individual-difference measures

	IH_score	AOT_score	IC_HI	IC_VI	IC_HC	IC_VC
IH_score	1.00	0.36	0.09	-0.23	0.37	0.10
AOT_score	0.36	1.00	0.00	-0.28	0.11	-0.20
IC_HI	0.09	0.00	1.00	0.28	0.05	0.15
IC_VI	-0.23	-0.28	0.28	1.00	-0.08	0.29
IC_HC	0.37	0.11	0.05	-0.08	1.00	0.35
IC_VC	0.10	-0.20	0.15	0.29	0.35	1.00

Table 34: Model for H1, including all individual differences measures as predictors.

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	0.262	0.201	[-0.133, 0.656]	.193
feedback	0.345	0.046	[0.255, 0.436]	< .001
justification	0.252	0.041	[0.172, 0.332]	< .001
IH_score	0.046	0.036	[-0.024, 0.116]	.195
AOT_score	0.154	0.033	[0.089, 0.219]	< .001
IC_HI	-0.004	0.032	[-0.066, 0.058]	.897
IC_VI	-0.050	0.034	[-0.117, 0.018]	.148
IC_HC	-0.085	0.035	[-0.154, -0.015]	.017
IC_VC	-0.029	0.035	[-0.098, 0.040]	.408
feedback:justification	-0.054	0.057	[-0.166, 0.059]	.349
feedback:IH_score	0.000	0.048	[-0.093, 0.094]	.993
feedback:AOT_score	-0.043	0.045	[-0.130, 0.045]	.338
feedback:IC_HI	0.025	0.043	[-0.058, 0.109]	.556
feedback:IC_VI	-0.001	0.046	[-0.091, 0.090]	.990
feedback:IC_HC	0.013	0.047	[-0.079, 0.106]	.775
feedback:IC_VC	0.050	0.047	[-0.042, 0.142]	.291
justification:IH_score	-0.008	0.048	[-0.102, 0.087]	.874
justification:AOT_score	-0.079	0.046	[-0.168, 0.010]	.084
justification:IC_HI	0.027	0.044	[-0.058, 0.112]	.536
justification:IC_VI	-0.006	0.047	[-0.098, 0.086]	.903
justification:IC_HC	0.043	0.048	[-0.050, 0.137]	.365
justification:IC_VC	0.000	0.047	[-0.093, 0.093]	.997
feedback:justification:IH_score	0.001	0.066	[-0.129, 0.132]	.985
feedback:justification:AOT_score	0.051	0.063	[-0.072, 0.174]	.418
feedback:justification:IC_HI	0.026	0.060	[-0.091, 0.143]	.664
feedback:justification:IC_VI	-0.002	0.065	[-0.129, 0.125]	.975
feedback:justification:IC_HC	-0.009	0.065	[-0.137, 0.120]	.895
feedback:justification:IC_VC	-0.038	0.065	[-0.165, 0.089]	.557
SD (Country: (Intercept))	0.421			
SD (Country: feedback)	0.114			
SD (Country: justification)	0.000			
SD (Country: feedback:justification)	0.057			
SD (problem: (Intercept))	0.464			
N	24218			
N Country	53			
N problem	6			

Table 35: Model for H2-H3, including all individual differences measures as predictors.

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	-0.510	0.226	[-0.953, -0.067]	.024
feedback	0.647	0.031	[0.587, 0.707]	< .001
justification	0.420	0.038	[0.346, 0.495]	< .001
IH_score	-0.051	0.024	[-0.098, -0.003]	.036
AOT_score	0.295	0.023	[0.250, 0.341]	< .001
IC_HI	-0.070	0.021	[-0.112, -0.028]	.001
IC_VI	0.047	0.023	[0.001, 0.092]	.044
IC_HC	-0.087	0.024	[-0.133, -0.040]	< .001
IC_VC	-0.099	0.023	[-0.144, -0.053]	< .001
feedback:justification	-0.043	0.048	[-0.137, 0.052]	.375
feedback:IH_score	0.049	0.033	[-0.014, 0.113]	.129
feedback:AOT_score	-0.074	0.031	[-0.135, -0.013]	.017
feedback:IC_HI	0.029	0.029	[-0.028, 0.086]	.326
feedback:IC_VI	0.013	0.032	[-0.049, 0.075]	.673
feedback:IC_HC	0.061	0.032	[-0.002, 0.124]	.056
feedback:IC_VC	0.066	0.032	[0.004, 0.128]	.038
justification:IH_score	-0.000	0.033	[-0.066, 0.065]	.994
justification:AOT_score	-0.019	0.032	[-0.082, 0.043]	.550
justification:IC_HI	0.045	0.030	[-0.014, 0.103]	.135
justification:IC_VI	0.030	0.032	[-0.034, 0.093]	.355
justification:IC_HC	0.053	0.033	[-0.012, 0.117]	.110
justification:IC_VC	-0.015	0.032	[-0.079, 0.048]	.631
feedback:justification:IH_score	-0.020	0.047	[-0.111, 0.071]	.669
feedback:justification:AOT_score	0.099	0.045	[0.012, 0.187]	.026
feedback:justification:IC_HI	-0.015	0.042	[-0.096, 0.067]	.725
feedback:justification:IC_VI	-0.040	0.045	[-0.129, 0.048]	.372
feedback:justification:IC_HC	-0.009	0.046	[-0.099, 0.080]	.837
feedback:justification:IC_VC	-0.051	0.045	[-0.140, 0.037]	.257
SD (Country: (Intercept))	0.439			
SD (Country: feedback)	0.066			
SD (Country: justification)	0.148			
SD (Country: feedback:justification)	0.153			
SD (problem: (Intercept))	0.530			
N	48036			
N Country	53			
N problem	6			

Individual Differences and Corrective Behaviour (revised)

In this version all scales are centred within country, to mitigate issues of violation of scalar invariance and improve comparability across countries.

Table 36: Model for H1, including all individual differences measures as predictors.

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	0.236	0.204	[-0.163, 0.636]	.246
feedback	0.349	0.045	[0.261, 0.437]	< .001
justification	0.251	0.041	[0.171, 0.332]	< .001
IH_score_wc	0.045	0.035	[-0.024, 0.115]	.200
AOT_score_wc	0.136	0.032	[0.073, 0.199]	< .001
IC_HI_wc	0.003	0.031	[-0.058, 0.065]	.920
IC_VI_wc	-0.042	0.033	[-0.107, 0.022]	.199
IC_HC_wc	-0.084	0.035	[-0.152, -0.015]	.017
IC_VC_wc	-0.025	0.034	[-0.092, 0.042]	.467
feedback:justification	-0.053	0.057	[-0.165, 0.059]	.351
feedback:IH_score_wc	0.004	0.048	[-0.089, 0.097]	.932
feedback:AOT_score_wc	-0.032	0.044	[-0.118, 0.053]	.460
feedback:IC_HI_wc	0.015	0.042	[-0.068, 0.098]	.724
feedback:IC_VI_wc	-0.001	0.044	[-0.088, 0.086]	.975
feedback:IC_HC_wc	0.016	0.046	[-0.075, 0.107]	.732
feedback:IC_VC_wc	0.053	0.046	[-0.036, 0.142]	.245
justification:IH_score_wc	-0.008	0.048	[-0.102, 0.087]	.876
justification:AOT_score_wc	-0.074	0.045	[-0.162, 0.014]	.099
justification:IC_HI_wc	0.012	0.043	[-0.073, 0.097]	.776
justification:IC_VI_wc	-0.000	0.046	[-0.089, 0.089]	.999
justification:IC_HC_wc	0.046	0.048	[-0.047, 0.139]	.333
justification:IC_VC_wc	-0.009	0.046	[-0.100, 0.081]	.841
feedback:justification:IH_score_wc	0.001	0.067	[-0.129, 0.132]	.985
feedback:justification:AOT_score_wc	0.043	0.062	[-0.077, 0.164]	.482
feedback:justification:IC_HI_wc	0.050	0.059	[-0.066, 0.166]	.401
feedback:justification:IC_VI_wc	-0.007	0.063	[-0.130, 0.116]	.909
feedback:justification:IC_HC_wc	-0.017	0.065	[-0.145, 0.110]	.789
feedback:justification:IC_VC_wc	-0.022	0.063	[-0.146, 0.103]	.734
SD (Country: (Intercept))	0.475			
SD (Country: feedback)	0.094			
SD (Country: justification)	0.000			
SD (Country: feedback:justification)	0.052			
SD (problem: (Intercept))	0.464			
N	24218			
N Country	53			
N problem	6			

Table 37: Model for H2-H3, including all individual differences measures as predictors.

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	-0.545	0.229	[-0.995, -0.095]	.017
feedback	0.646	0.030	[0.586, 0.705]	< .001
justification	0.415	0.038	[0.341, 0.489]	< .001
IH_score_wc	-0.055	0.024	[-0.101, -0.008]	.022
AOT_score_wc	0.274	0.022	[0.230, 0.317]	< .001
IC_HI_wc	-0.065	0.021	[-0.106, -0.024]	.002
IC_VI_wc	0.037	0.022	[-0.006, 0.080]	.093
IC_HC_wc	-0.079	0.023	[-0.125, -0.033]	< .001
IC_VC_wc	-0.088	0.023	[-0.133, -0.044]	< .001
feedback:justification	-0.047	0.049	[-0.142, 0.049]	.338
feedback:IH_score_wc	0.055	0.032	[-0.008, 0.119]	.089
feedback:AOT_score_wc	-0.064	0.030	[-0.124, -0.005]	.034
feedback:IC_HI_wc	0.023	0.029	[-0.034, 0.079]	.430
feedback:IC_VI_wc	0.023	0.031	[-0.037, 0.083]	.453
feedback:IC_HC_wc	0.056	0.032	[-0.006, 0.118]	.078
feedback:IC_VC_wc	0.056	0.031	[-0.005, 0.117]	.070
justification:IH_score_wc	0.010	0.033	[-0.055, 0.075]	.765
justification:AOT_score_wc	-0.013	0.031	[-0.073, 0.048]	.679
justification:IC_HI_wc	0.043	0.029	[-0.014, 0.101]	.141
justification:IC_VI_wc	0.038	0.031	[-0.023, 0.099]	.223
justification:IC_HC_wc	0.043	0.032	[-0.021, 0.106]	.188
justification:IC_VC_wc	-0.020	0.031	[-0.082, 0.041]	.518
feedback:justification:IH_score_wc	-0.029	0.046	[-0.120, 0.061]	.525
feedback:justification:AOT_score_wc	0.079	0.043	[-0.006, 0.164]	.069
feedback:justification:IC_HI_wc	-0.010	0.041	[-0.090, 0.071]	.816
feedback:justification:IC_VI_wc	-0.049	0.044	[-0.135, 0.036]	.260
feedback:justification:IC_HC_wc	-0.001	0.045	[-0.089, 0.087]	.984
feedback:justification:IC_VC_wc	-0.037	0.044	[-0.124, 0.049]	.398
SD (Country: (Intercept))	0.519			
SD (Country: feedback)	0.061			
SD (Country: justification)	0.146			
SD (Country: feedback:justification)	0.160			
SD (problem: (Intercept))	0.531			
N	48036			
N Country	53			
N problem	6			

Community and Student Sample Differences

```
# to check
table(d$Sample)
```

```
234455      5 Chttp:// Community Student
      1      4      2      4334    10669
```

```
sum(is.na(d$Sample))
```

```
[1] 235
```

```
# unique(d$Country[is.na(d$Sample)])
# unique(d$LabID[is.na(d$Sample)])
```

```
d %>%
  filter(is.na(Sample)) %>%
  group_by(Country, LabID) %>%
  summarise(N = n())
```

```
# A tibble: 11 x 3
# Groups:   Country [9]
  Country    LabID      N
  <chr>      <chr>  <int>
1 Bangladesh <NA>         3
2 Colombia   3629        29
3 Germany    Sample=S     1
4 Germany    <NA>         1
5 Italy      3704...      1
6 Italy      <NA>        39
7 Mexico     15          1
8 Netherlands 381        77
9 Norway     <NA>        79
10 Serbia    <NA>         3
11 Spain     3275         1
```

Table 38: Model for H1, including sample type (Community vs. Student) as fixed-effect.

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	0.137	0.209	[-0.272, 0.547]	.511
feedback	0.336	0.071	[0.197, 0.474]	< .001
justification	0.207	0.072	[0.065, 0.349]	.004
SampleStudent	0.172	0.067	[0.040, 0.304]	.011
feedback:justification	-0.100	0.099	[-0.295, 0.095]	.313
feedback:SampleStudent	0.024	0.087	[-0.147, 0.195]	.784
justification:SampleStudent	0.066	0.088	[-0.106, 0.238]	.452
feedback:justification:SampleStudent	0.094	0.121	[-0.142, 0.330]	.436
SD (Country: (Intercept))	0.471			

Predictor	Estimate	SE	95% CI	p-value
SD (Country: feedback)	0.076			
SD (Country: justification)	0.000			
SD (Country: feedback:justification)	0.042			
SD (problem: (Intercept))	0.468			
N	23760			
N Country	52			
N problem	6			

Table 39: Model for H2-H3, including sample type (Community vs. Student) as fixed-effect.

Predictor	Estimate	SE	95% CI	p-value
(Intercept)	-0.573	0.226	[-1.015, -0.131]	.011
feedback	0.703	0.053	[0.600, 0.806]	< .001
justification	0.440	0.059	[0.325, 0.556]	< .001
SampleStudent	0.041	0.049	[-0.054, 0.136]	.400
feedback:justification	-0.162	0.080	[-0.318, -0.006]	.042
feedback:SampleStudent	-0.091	0.062	[-0.212, 0.030]	.142
justification:SampleStudent	-0.033	0.067	[-0.164, 0.098]	.622
feedback:justification:SampleStudent	0.155	0.092	[-0.025, 0.336]	.092
SD (Country: (Intercept))	0.513			
SD (Country: feedback)	0.048			
SD (Country: justification)	0.152			
SD (Country: feedback:justification)	0.167			
SD (problem: (Intercept))	0.522			
N	47290			
N Country	52			
N problem	6			

Confidence Calibration and Intellectual Humility

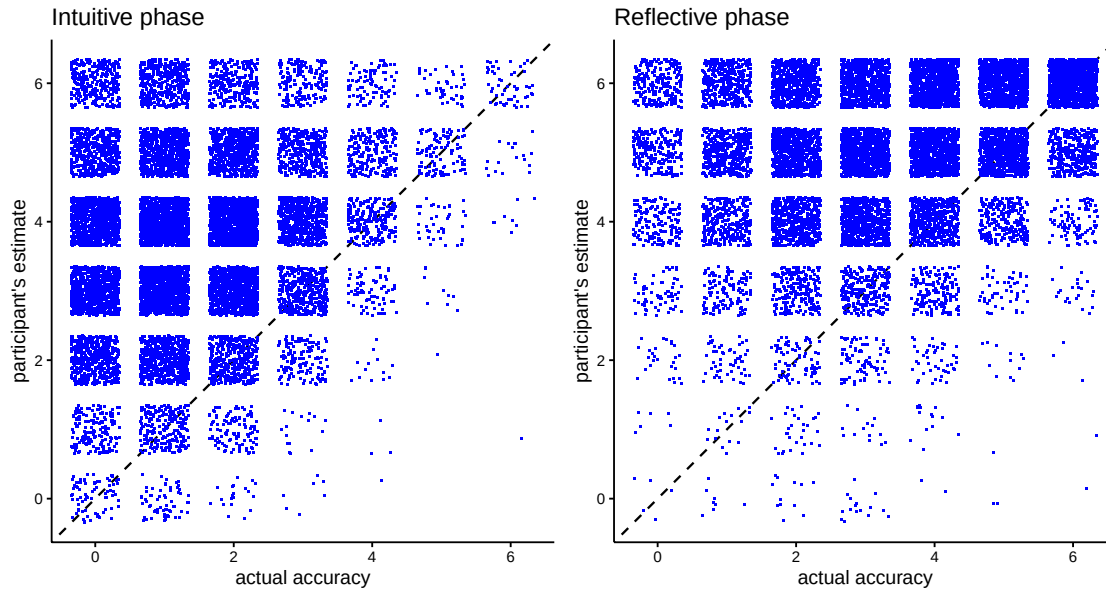


Figure 2: Number of correct answer (vertical coordinates) plotted as a function of the answer to the calibration responses (e.g. the participant's estimates of their own accuracy; horizontal coordinates). Some jitter added for visibility. Both plots demonstrates substantial overconfidence, since in both cases most points lie below the identity line. For this plot we only excluded problems where participants declared using external resources.

Table 40: Generalised additive models predicting participants' estimates of performance.

Model	Component	Term	Estimate	SE	edf	Ref. df	Statistic	p
Intuitive	Parametric	Intercept	3.725	0.011			349.164	< .001
Intuitive	Smooth	Actual accuracy			4.121	4.813	151.728	< .001
Intuitive	Smooth	Intellectual humility			2.279	2.757	10.036	< .001
Intuitive	Smooth	Actual accuracy × intellectual humility			1.894	2.388	0.866	.510
Reflective	Parametric	Intercept	4.975	0.009			583.248	< .001
Reflective	Smooth	Actual accuracy			4.804	5.518	298.650	< .001
Reflective	Smooth	Intellectual humility			1.997	2.456	1.202	.281
Reflective	Smooth	Actual accuracy × intellectual humility			1.622	2.019	4.232	.014

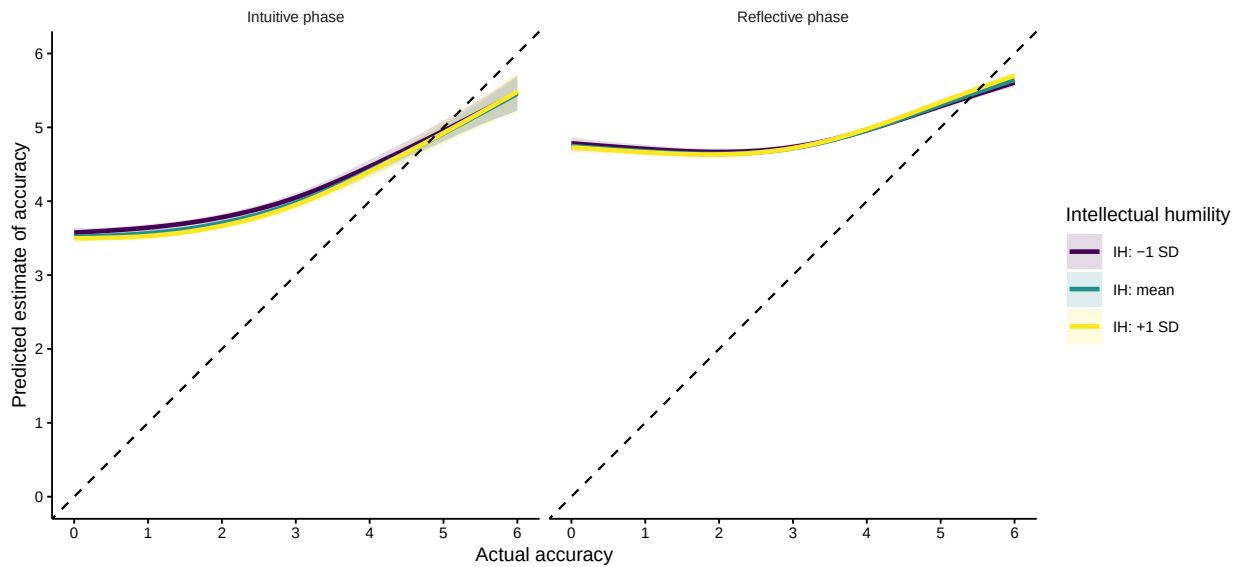


Figure 3: Intellectual humility and calibration of estimated performance. Generalised additive models (GAMs) predicting participants' estimated number of correct answers from their actual accuracy and intellectual humility, fitted separately for the intuitive (left) and reflective (right) phases. Lines show model-predicted estimates for intellectual humility at the mean and ± 1 SD; shaded bands indicate 95% confidence intervals. The dashed diagonal indicates perfect calibration, such that values above the line reflect overestimation and values below the line reflect underestimation. Intellectual humility was associated with a small overall shift in estimates in the intuitive phase, whereas in the reflective phase it showed a small but significant interaction with actual accuracy.

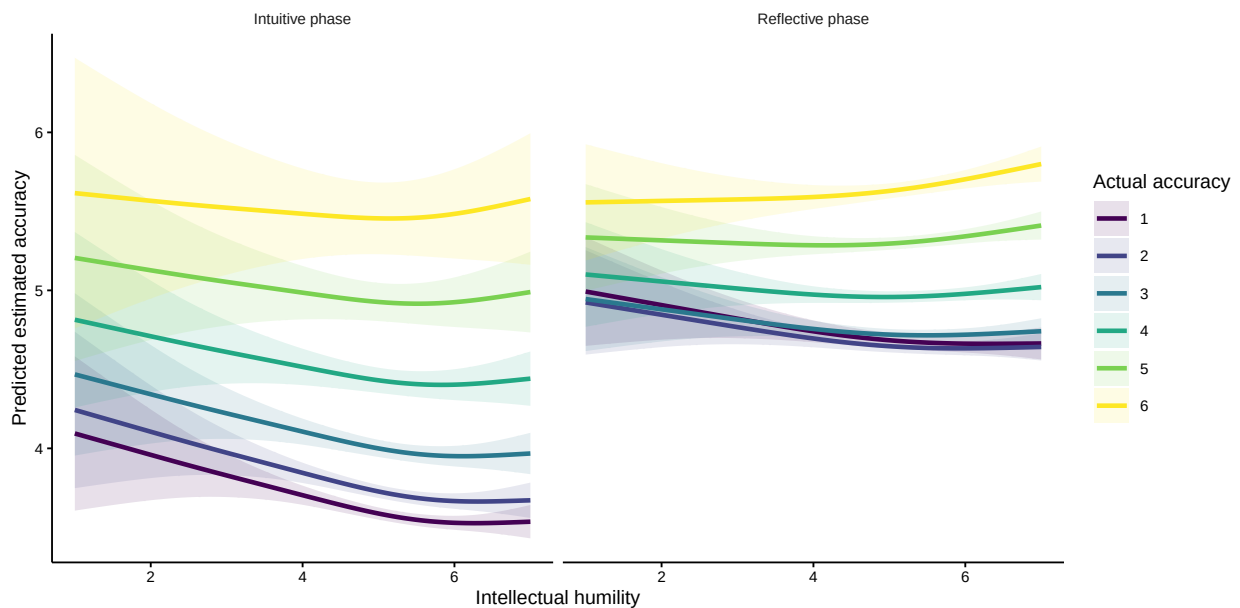


Figure 4: Intellectual humility, actual accuracy, and estimated performance. Generalised additive models (GAMs) predicting participants' estimated number of correct answers from actual accuracy and intellectual humility, fitted separately for the intuitive (left) and reflective (right) phases. Lines show model-predicted estimates across the observed range of intellectual humility scores, plotted separately for each level of actual accuracy; shaded bands indicate 95% confidence intervals. In the intuitive phase, intellectual humility was associated with a small overall downward shift in estimated performance across accuracy levels. In the reflective phase, the association between intellectual humility and estimated performance varied modestly as a function of actual accuracy, consistent with a small but significant interaction.

Robustness analysis: quasibinomial GAMs

As a robustness analysis, we repeated the calibration analyses using quasibinomial GAMs. These models constrain predicted estimates to the possible 0–6 range while allowing for overdispersion relative to the binomial variance.

Model estimates

Table 41: Quasibinomial generalised additive models predicting participants' estimates of performance.

Model	Component	Term	Estimate	SE	edf	Ref. df	Statistic	p
Intuitive	Parametric	Intercept	0.502	0.008			65.621	< .001
Intuitive	Smooth	Actual accuracy			4.420	5.032	135.304	< .001
Intuitive	Smooth	Intellectual humility			2.187	2.661	10.229	< .001
Intuitive	Smooth	Actual accuracy × intellectual humility			1.610	2.009	0.582	.570
Reflective	Parametric	Intercept	1.662	0.011			150.685	< .001
Reflective	Smooth	Actual accuracy			5.151	5.754	257.390	< .001
Reflective	Smooth	Intellectual humility			2.011	2.473	2.162	.093
Reflective	Smooth	Actual accuracy × intellectual humility			2.341	2.936	5.344	.002

Predicted calibration curves

Intellectual humility interaction plots

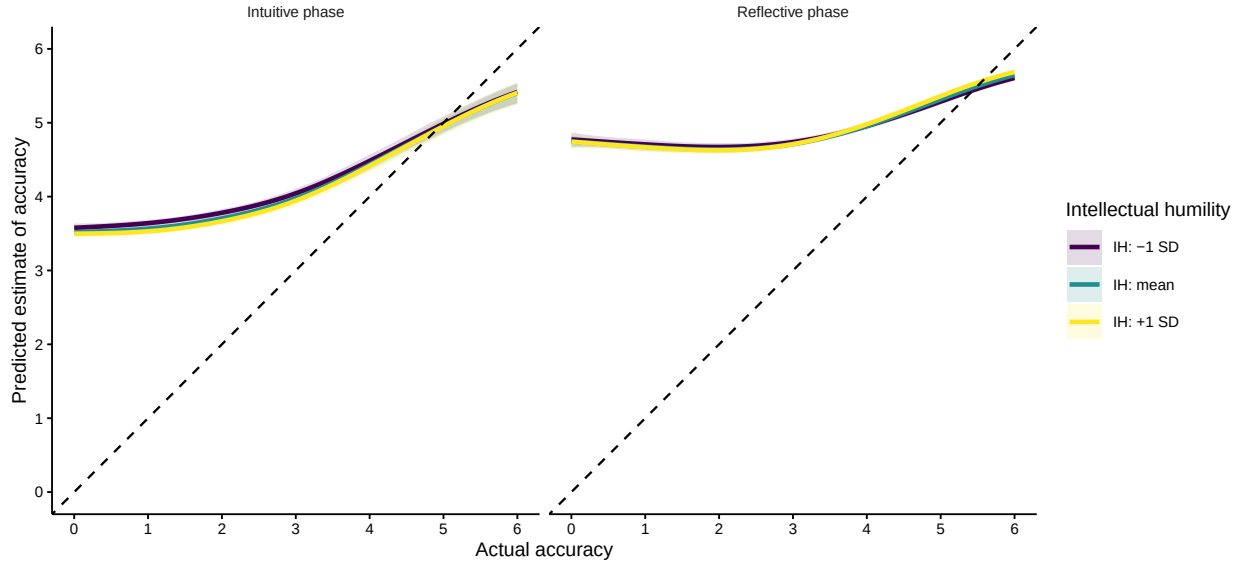


Figure 5: Quasibinomial robustness analysis of intellectual humility and calibration of estimated performance. Lines show model-predicted estimates at the mean and ± 1 SD of intellectual humility; shaded bands indicate 95% confidence intervals. The dashed diagonal indicates perfect calibration.

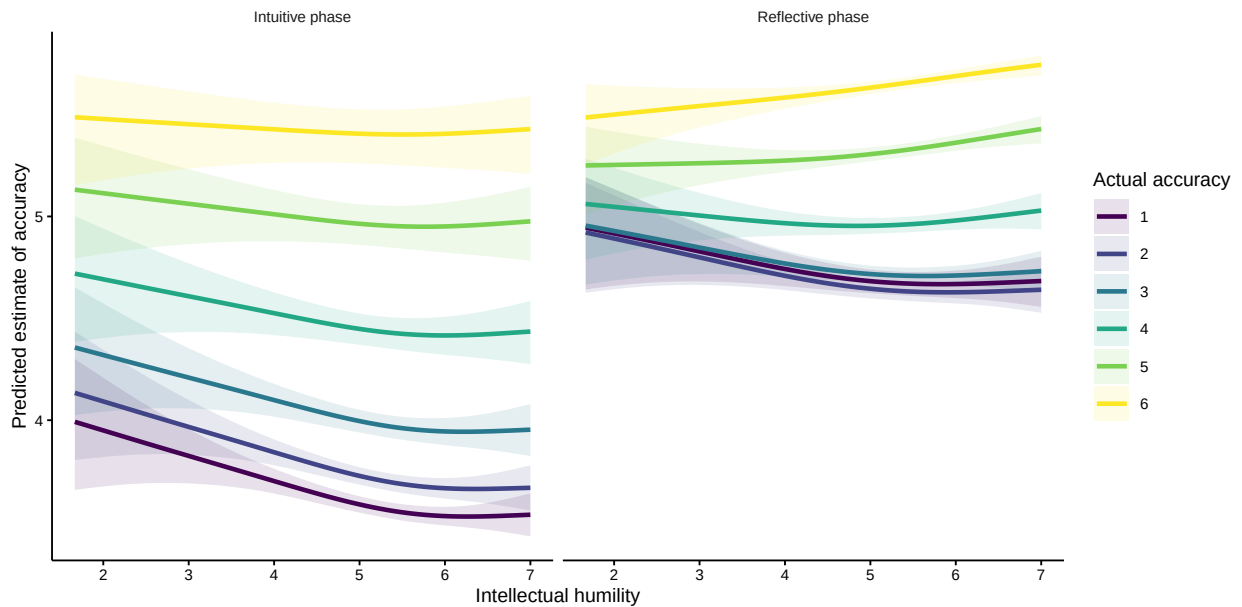


Figure 6: Quasibinomial robustness analysis of intellectual humility, actual accuracy, and estimated performance. Lines show model-predicted estimates across the observed range of intellectual humility scores, separately for each level of actual accuracy; shaded bands indicate 95% confidence intervals.